



MICROSOFT ENTRA ID · OPERATIONAL RUNBOOK

# SMS & Voice MFA Registration Report

**Script** `Get-SmsVoiceMfaUsers.ps1`**Purpose** Identify every user in a Microsoft Entra ID tenant who has a phone number registered as an authentication method, ahead of the Microsoft-provided SMS and Voice retirement on \*\*1 February 2027\*\*.**Access required** Read-only. The script issues GET requests only and never requests a write scope.

## 1. Before you start

| REQUIREMENT | DETAIL                                                                                                   |
|-------------|----------------------------------------------------------------------------------------------------------|
| PowerShell  | 5.1 or PowerShell 7.x                                                                                    |
| Module      | <code>Microsoft.Graph.Authentication</code> 2.0.0 or later                                               |
| Entra role  | Global Reader, Security Reader, or Reports Reader                                                        |
| Network     | Outbound HTTPS to <code>graph.microsoft.com</code>                                                       |
| Time to run | 1–2 minutes for a standard run; add ~1 minute per 500 users if <code>-IncludePhoneNumbers</code> is used |

**Install the module (one time):**

```
Install-Module Microsoft.Graph.Authentication -MinimumVersion 2.0.0 -Scope CurrentUser
```

If the tenant blocks PSGallery, install from the organisation's internal repository instead.

## 2. Permissions to approve

The script asks for consent to the following delegated scopes. Share this table with whoever approves the consent prompt.

| SCOPE                                          | WHEN REQUESTED                              | WHY                                                                                                       |
|------------------------------------------------|---------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| <code>AuditLog.Read.All</code>                 | Always                                      | Reads <code>/reports/authenticationMethods/userRegistrationDetails</code> — the registered-methods report |
| <code>UserAuthenticationMethod.Read.All</code> | Only with <code>-IncludePhoneNumbers</code> | Reads each affected user's phone method to return the number                                              |

No write, directory-modify, or mail scope is ever requested. If an admin consent prompt shows anything beyond the two scopes above, stop and investigate.

## 3. Run it

**Standard run** — produces both a CSV and an HTML report in the current folder:

```
.\Get-SmsVoiceMfaUsers.ps1
```

**Recommended for a client engagement** — pin the tenant explicitly and name the output:

```
.\Get-SmsVoiceMfaUsers.ps1 -TenantId contoso.onmicrosoft.com `
                          -CustomerName "Contoso Ltd" `
                          -OutputFolder C:\Engagements\Contoso
```

**Always pass `-TenantId` when you work across multiple tenants.** It is what stops the script reusing a Graph session left over from a previous client. Without it, the script will use whatever session is already open.

A browser window opens for sign-in. Sign in with an account holding one of the roles listed in section 1.

### Parameters

| PARAMETER                         | EFFECT                                                                    |
|-----------------------------------|---------------------------------------------------------------------------|
| <code>-TenantId</code>            | Tenant GUID or domain. Validates the session belongs to the right tenant. |
| <code>-CustomerName</code>        | Organisation name shown in the HTML header. Auto-detected if omitted.     |
| <code>-OutputFolder</code>        | Where files are written. Default: current directory.                      |
| <code>-OutputFormat</code>        | CSV , HTML , or Both . Default: Both .                                    |
| <code>-IncludePhoneNumbers</code> | Adds masked phone numbers. Slower, needs the extra scope.                 |
| <code>-ExcludeGuests</code>       | Omits B2B guest accounts.                                                 |
| <code>-KeepSession</code>         | Leaves the Graph session connected after the run.                         |

## 4. Read the output

### Console summary

```
Users checked                4,182
USERS WITH SMS OR VOICE REGISTERED 240
  of which can receive SMS      87
  privileged accounts          14
NO OTHER MFA METHOD - act first  74
```

The last line is the number that drives the remediation plan.

### Priority column

Every affected user is assigned one of three priorities. The CSV is sorted by it, so the top of the file is the work queue.

| PRIORITY | MEANING                                            | WHAT TO DO                                                                                                                                |
|----------|----------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| P1       | Phone is the user's only MFA method                | Register an alternative (passkey or Authenticator) <b>before</b> removing the phone. These users are blocked at sign-in after 1 Feb 2027. |
| P2       | Has an alternative, but phone is still the default | Change the default method, then remove the number.                                                                                        |
| P3       | Alternative already in place and already default   | Remove the phone number. No user impact.                                                                                                  |

## Files produced

| FILE                                                   | AUDIENCE              | USE                                                                        |
|--------------------------------------------------------|-----------------------|----------------------------------------------------------------------------|
| <a href="#">SMS-Voice-MFA-Users-&lt;date&gt;.csv</a>   | Operations / helpdesk | Working list, filter and assign                                            |
| <a href="#">SMS-Voice-MFA-Report-&lt;date&gt;.html</a> | Management / CISO     | Self-contained report; open in any browser, <a href="#">Ctrl+P</a> for PDF |

The HTML report has a search box, priority filters, sortable columns, and an **Export current view to CSV** button — filter to P1, export, and that is the helpdesk queue.

## Understanding the SMS and Voice columns

Entra does not store SMS and Voice as two separate registrations. It stores a phone number with a type:

| REGISTERED TYPE  | CAN RECEIVE SMS | CAN RECEIVE VOICE CALL |
|------------------|-----------------|------------------------|
| Mobile           | Yes             | Yes                    |
| Alternate mobile | No              | Yes                    |
| Office           | No              | Yes                    |

Every user in the report is in scope for the retirement, regardless of type.

## 5. Key dates

| DATE        | WHAT HAPPENS                                                                                                                    |
|-------------|---------------------------------------------------------------------------------------------------------------------------------|
| 1 Sep 2026  | Users enabled for SMS or Voice are auto-enabled for passkeys and prompted to register at next MFA. The prompt can be dismissed. |
| 18 Sep 2026 | Microsoft publishes supported third-party telecom providers.                                                                    |
| 30 Oct 2026 | Organisations still needing SMS or Voice must have a provider configured.                                                       |
| 1 Feb 2027  | Microsoft-provided SMS and voice stop working, including for SSPR.                                                              |

Note that the September auto-enablement targets users **enabled** for SMS or Voice in the Authentication Methods Policy — usually a larger set than the users who happen to have a number registered. Check the policy separately if you need that scope.

## 6. Troubleshooting

| SYMPTOM                                                                   | CAUSE AND FIX                                                                                                                                                                               |
|---------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| The registration report returned no records                               | Consent or role missing. Confirm <code>AuditLog.Read.All</code> is consented and the account holds Global Reader, Security Reader, or Reports Reader. This is an error, not a clean tenant. |
| Existing Graph session does not match this request. Reconnecting.         | Expected and correct. A session from another tenant was open; the script disconnected it.                                                                                                   |
| Microsoft.Graph.Authentication X is installed; 2.0.0 or later is required | Run <code>Update-Module Microsoft.Graph.Authentication</code> .                                                                                                                             |
| N of M phone-number lookups failed                                        | Missing <code>UserAuthenticationMethod.Read.All</code> , or throttling. User counts are still correct; only the number column is affected.                                                  |
| Script will not run — execution policy                                    | Preferred: use a signed copy. Temporary: <code>Set-ExecutionPolicy -Scope Process -ExecutionPolicy Bypass</code> . Do not change machine-wide policy.                                       |
| Counts shift slightly between runs                                        | The registration report is generated data and can lag the directory by a few hours. Normal during active remediation.                                                                       |

## 7. Handling the output

The CSV and HTML contain personal data: display names, user principal names, privileged-role status, and — if `-IncludePhoneNumbers` was used — masked phone numbers.

- Store in the engagement's approved location only.
- Transfer via the client's approved channel, not personal email or messaging.
- Apply the retention period in the engagement's data-protection terms; delete on closure.
- Numbers are masked by default. There is no unmask option in this version by design.

On Windows the script automatically restricts file permissions to the account that ran it.

## 8. Before standardising across clients

- ☐ Run end-to-end in a lab tenant, both with and without `-IncludePhoneNumbers`
- ☐ Verify the cross-tenant guard by running against two tenants in one session
- ☐ Run `Invoke-ScriptAnalyzer -Severity Warning,Error` and clear findings
- ☐ Test with an under-privileged account to confirm it fails loudly
- ☐ Code-sign the script and publish its SHA256 hash
- ☐ Have a second engineer review it

Data source: Microsoft Graph `/reports/authenticationMethods/userRegistrationDetails`.